

Cricket Crossword Solutions

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1 3	2 5
3 6	6

ONE ACROSS

This is a standard problem of choosing k non-adjacent items from n items, given by

$$\binom{n-k+1}{k}.$$

Here, $n = 9$ and $k = 3$. Thus,

$$\binom{9-3+1}{3} = \binom{7}{3} = \boxed{35}.$$

ONE DOWN

We distribute 4 distinct awards among 3 franchises such that each franchise gets at least one award.

Total ways without restriction:

$$3^4 = 81.$$

Using inclusion-exclusion:

The number of ways where one franchise gets nothing is

$$\binom{3}{1} \cdot 2^4 = 3 \cdot 16 = 48,$$

and the number of ways where two franchises get nothing is

$$\binom{3}{2} \cdot 1^4 = 3.$$

Thus, the required number of ways turns out to be

$$81 - 48 + 3 = \boxed{36}.$$

TWO DOWN

Let x_i denote the number of overs bowled by the i -th bowler. Then,

$$x_1 + x_2 + x_3 + x_4 + x_5 + x_6 = 9$$

with

$$1 \leq x_i \leq 5.$$

To remove the lower bound, let

$$y_i = x_i - 1.$$

Then,

$$y_1 + y_2 + y_3 + y_4 + y_5 + y_6 = 3,$$

with

$$0 \leq y_i \leq 4.$$

Now, apply stars and bars (ignoring the upper bound temporarily):

$$\binom{3+6-1}{6-1} = \binom{8}{5} = \boxed{56}.$$

Since the total sum is only 3, the upper bound $y_i \leq 4$ is never violated.

THREE ACROSS

We want to count the number of ways to color a cycle of length n using k colors such that no two adjacent vertices have the same color.

The formula for this is

$$(k-1)^n + (-1)^n(k-1)$$

If you're unfamiliar with this expression, then it helps to think of the first term as the number of valid arrangements, and the second term simply accounts for the overcounting.

For $n = 6$ and $k = 3$:

$$(3-1)^6 + (-1)^6(3-1) = 2^6 + 2 = 64 + 2 = \boxed{66}.$$